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Improving customer retention in taxi industry using travel data analytics: A churn prediction study

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ABSTRACT

The retention of public transport users is widely acknowledged as a paramount challenge in the path towards the establishment of more sustainable cities and societies. In this setting, in which no contractual relationship with customers exists, an early and accurate prediction of whether a customer will remain with the company or leave, assumes great significance for businesses to develop effective retention strategies. This work focuses on this topic by identifying potential churners based on their past travel behavior. To achieve this, we developed a set of classification models using various machine learning techniques. These models were then employed as base learners within a stacking ensemble. All classifiers were developed with a profit-driven approach, optimizing for expected maximum profit. Finally, we calculated Shapley Additive Explanation values to enhance the interpretability of the proposed classifiers. The performance of the predictive models was evaluated using the data of taxi services recorded in a Portuguese city for 52 months. A broad range of predictors is proposed, including recency and frequency measures of taxi usage as well as others related to customers' satisfaction level. The predictive power of the models was also assessed for specific proportions of higher risk customers. All models have shown the capability to identify churners accurately. This study innovates in evaluating the one-to-one service provider company-customer relationship in the context of taxi industry. Retention actions to promote customers loyalty and enhance retention are also suggested.

1. Introduction

Increasing transit ridership is a top priority for practitioners in the transportation sector, including cities authorities and transport operator managers, seeking a decrease in the use of private cars and traffic in urban areas (Minser and Webb, 2010; Saleh and Farrell, 2007). Such actions are a crucial step towards the establishment of more sustainable transport systems within cities, decreasing the usage of private vehicles and increasing the attractiveness of public transport. The shift towards a more efficient transport network can reduce pollution emissions and improves the quality of life of those living in cities (Lee and Erickson, 2017). Such a matter requires improved efforts and strategies from transport operators', focused as much on keeping current riders as on attracting new ones. In the highly competitive setting of transportation, the proliferation of competitors has made the latter an increasingly challenging task. Therefore, in the past few years, companies have been devoting deeper efforts to strengthen the relationship with current riders, increasing their loyalty (Carrel and Li, 2019; Habib et al., 2019; Lee et al., 2011). In the context of public transit, customer retention is colloquially known as transit retention and is related to the likelihood of a rider continuing to use public transport over time (Habib et al., 2019; Huan et al., 2022).

The benefits of customer retention have been pointed out by several authors, who emphasize the lower sensitivity of loyal customers to the service/product price and to the competitors' marketing campaigns (Lariviere and Van den Poel, 2005; Liu et al., 2024b; Miguéis et al., 2012; Minser and Webb, 2010). Moreover, loyal customers may gradually acquire more goods and services over time and are associated with reduced marketing costs, as they indirectly attract new customers through positive word-of-mouth. For those reasons, higher customer retention rates

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result in increases in company profits (Maldonado et al., 2020; Miguéis et al., 2013; Silveira et al., 2022).

Anticipating customer attrition, or churn, is crucial for retention, allowing businesses to implement targeted strategies that strengthen relationships and enhance engagement (Lee et al., 2011; Wen et al., 2022). An accurate disclosure of churners can be even more decisive for businesses in which customers do not need to communicate in advance their intention to quit, and the costs of switching to other product/service provider are low or non-existent, representing a key driver for the competitiveness of these companies.

Due to its acknowledged relevance in the marketing strategic plan of companies operating in the most diverse activity sectors, Customer churn prediction (CCP) has been widely explored in the literature. Banking and financial services (Elyusufi and Ait Kbir, 2022; Lariviere and Van den Poel, 2005), retail (Benchekroun et al., 2022; Miguéis et al., 2013) and telecommunication (Adebiyi et al., 2016; Lee et al., 2011; Slof et al., 2021; Wen et al., 2022) are domains in which this topic has been receiving particular attention from researchers. However, research targeting to foresee the abandonment of customers in transport setting is much less remarkable, with few studies being found in the literature. The recent works of Hu et al. (2021) and Park et al. (2022) are two examples of such studies, both of which are the target of deeper discussion in Section 2.

Studies supporting these sectors have evolved to prioritize the prediction of customer churn based on profitability metrics, such as the *Expected Maximum Profit Criterion* (EMPC), introduced by Verbraken et al. (2013). The adoption of profit-driven metrics has led to the development of predictive models aimed at maximizing overall revenue and minimizing the costs associated with customer acquisition. The adoption of EMPC by researchers such as Hoeppner et al. (2020), Maldonado et al. (2020), and Jiang et al. (2024), as well as its adaptation to different domains as demonstrated by Liu et al. (2025), highlights its value in identifying the most profitable churn prediction models and optimizing customer retention strategies. In this research work, we concentrate on EMPC, as it mirrors scenarios more commonly encountered in real-world applications.

Finally, due to the inherent opacity of black-box models like machine learning and ensemble methods, post-hoc interpretability analyses are commonly applied, aiming to enhance transparency and understand the rationale behind predictions made by those models. For that purpose, Shapley Additive eXplanations (SHAP), developed by Lundberg and Lee (2017), has emerged as a leading framework for comprehensive model interpretability, allowing to effectively reveal feature contributions and their impact on model predictions. Thus, post-hoc interpretability analyses are nowadays commonly applied to gain insights into the model's decision-making process. In the present work, we seek the early identification of potential churners informed by the travel behavior of customers. For that purpose, we apply machine learning techniques and compare the performance of such methods. To enhance model robustness and accuracy, we explore the use of an ensemble stacking approach, combining predictions from multiple base classifiers. Furthermore, we focus on the model's interpretability by utilizing SHAP, a post-hoc method that effectively elucidates the contribution of each feature to individual and/or global predictions. Overall, this study intends to fill the gap of predicting customer churn in taxi industry, that is still an unexplored research field. Our findings can support taxi providers in developing enhanced retention strategies aiming to mitigate customer attrition at an individual level.

The structure of the remainder of this paper is as follows. In Section 2, we review previous research on transit user loyalty towards public transport and churn prediction modeling. Then, in Section 3, we describe in detail the modeling methodology we follow, namely the evaluation criteria and metrics and the classification techniques. In Section 4, we present a detailed description of the data, with a special focus on the predictive variables. In Section 5, we summarise and discuss the results of our analysis. Lastly, in Section 6, we draw the conclusions and present suggestions for future work.

2. Related work

In this section, we first revise prior research on transit user loyalty with respect to public transport (Section 2.1). Then, existing literature in customer churn prediction is addressed in Section 2.2.

2.1. Transit user loyalty

Customer loyalty is strongly linked to customer satisfaction, with several authors claiming customer satisfaction as the main driver of customer loyalty (e.g. Imaz et al. (2015); Minser and Webb (2010); Molina et al. (2021)). Customer satisfaction is defined as an evaluation process in which the customer's perceptions resultant of actual experience and his/her expectations regarding the product/service are compared (Park et al., 2022; Sukhov et al., 2023). Concerning the transport sector, when the travel experience meets customers' expectations, customer satisfaction takes place. In this specific domain, the frequency of using a particular mode of transportation is commonly studied to identify the customers' propensity to adopt a transport mode, working as a measure of customer loyalty towards that mode of transportation. (Fu, 2022; Wang et al., 2022; Yang et al., 2019).

An effective awareness of which aspects have the greatest impact on transit customer loyalty is crucial to support practitioners and policymakers in defining strategies that successfully boost the retention of riders and decrease ridership churn (Carrel and Li, 2019; Fu, 2022; van Lierop et al., 2018). Therefore, numerous research works investigating the factors affecting transit users' loyalty towards different modes of public transport have emerged in the literature. Zhao et al. (2014) (bus and train), Imaz et al. (2015) (bus and subway), Efthymiou et al. (2018) (bus), Habib et al. (2019) (train), Le et al. (2020) (multiple modes), Vicente et al. (2020) (multiple modes) and Si et al. (2022) (ride-sharing) are representative of this growing body of the literature. A consistent finding in the literature is that operational (frequency and punctuality), customer relation (kindness and helpfulness of operators) and on-board experience aspects (safety, cleanliness and comfort) are the features most tightly related to riders' loyalty. For a deeper and extended analysis of this branch of the literature, the reader is referred to van Lierop et al. (2018), Habib et al. (2019) and Silveira et al. (2022).

A notable point is the overwhelming use of surveys as the data collecting tool employed, with most of these surveys being launched at a single point in time. Accordingly, the majority of these studies do not represent a longitudinal analysis, not allowing to infer changes in riders' travel behavior over time (Le et al., 2020; Saleh and Farrell, 2007). In such surveys, riders are often asked about their intentions to continue using the service. Nevertheless, significant bias between riders' future intentions and their revealed travel behavior have been detected in recent investigations, leveraging the question about the reliability of users' previously stated intentions as indicator of their actual travel behavior (Carrel and Li, 2019; Le et al., 2020). Indeed, Carrel and Li (2019) note that studies based on data collected through surveys, may result in an overestimation of users' propensity to keep their level of ridership, and a consequent overestimation of users' loyalty and underestimation of users' retention. Those authors also argue that to get more accurate measures of transit users' loyalty, studies should rely on surveys launched at multiple points in time or use data collected from multiple sources (e.g. a survey and an automated fare collection system). Following this line of thought, Le et al. (2020) assessed the usage of satisfaction as a reliable predictor of rider attrition, using data collected from surveys at two points in time, to prevent errors caused by users' stated intentions. These authors have found a longitudinal relation between users' disappointment experiences and users' attrition. Not following that generalized tendency, the works of Trepanier et al. (2012), Ma et al. (2020)

and Chen et al. (2022) are among the few that explore smart card data to investigate what drives transit user loyalty.

Concerning the data analysed, most studies use a combination of demographic and socioeconomic data (e.g. gender, age, educational level, professional occupation, household income and size, car ownership), system usage data (e.g. frequency of use, how long user has been using public transport, transport modes used, availability of alternative modes) and trip characteristics (e.g. origin and destination, boarding and alighting time, trip purpose). Besides that, surveys can also include questions linked to service quality, customer satisfaction and problem experience dimensions. Depending on the scope and purpose of the study, respondents can also be asked about a particular mode of transport, such as the condition at the stations, driving behavior, information provided at stops, number of transfers and lines used, etc (Efthymiou et al., 2018; Habib et al., 2019; Zhao et al., 2014).

With regard to the data analysis, statistical methods are clearly the dominant tools applied. Structural equation modeling (Fu, 2022; Minser and Webb, 2010) and its further extensions Partial least squares structural equation modeling (Si et al., 2022) and Hierarchical linear model (Chen, 2016), Multinomial logit models (Le et al., 2020), Ordered logit models (Efthymiou et al., 2018) and Binary logistic regression (Imaz et al., 2015) are among the methods employed by researchers.

While a significant amount of literature has already been published on understanding the factors that affect transit loyalty, to the best of our knowledge, only Molina et al. (2021) addresses loyalty in the case of taxi users. This study explores the links between users' loyalty and their profiles, performing a comparison between taxi and private-hire driver service users.

2.2. Predicting customer churn

Due to the growing competition and the high costs required by mass marketing campaigns, companies operating in numerous sectors are starved for tools that allow them to identify customers who show a high propensity for churn effectively. Having identified the potential churners, enhanced marketing retention strategies can be defined and employed, ultimately yielding higher revenues for companies. The development of accurate churn prediction models, usually based on machine learning techniques, is at the root of that process (Lee et al., 2011; Liu et al., 2024a; Wen et al., 2022).

Machine learning research in CCP traditionally focuses on improving model accuracy through feature engineering and hyperparameter optimization (De Caigny et al., 2018). However, recent research emphasizes the importance of profit-driven approaches. Profit-driven models prioritize maximizing financial gain by directly incorporating the costs and benefits of customer retention into the model training process. Unlike traditional models that aim to minimize overall misclassification errors, profit-driven models seek to identify and target customers whose retention will yield the highest return on investment. This paradigm shift has led to the development of innovative approaches, such as profit-driven feature selection and profit-driven model training algorithms. For example, researchers have explored techniques such as profit-driven feature selection using support vector machines (Maldonado et al., 2015) and developed profit-driven models based on logistic regression (Stripling et al., 2018), decision trees (Hoeppner et al., 2020) and ensemble models resulting from a combination of multiple classifiers (Jiang et al., 2024). Moreover, profit-driven CCP modeling has explored the balance between the returns from identifying churners and the costs incurred from misclassifications. In this framework, Jiang et al. (2023) introduced a hybrid model designed to maximize returns while minimizing associated costs. These advancements demonstrate a significant shift in the focus of churn prediction research, moving beyond accuracy-based metrics to prioritize the financial outcomes of model predictions. For additional pertinent studies on profit-driven churn prediction, the reader is referred to the works Jiang et al. (2023) and Jiang et al. (2024).

As mentioned in Section 1, CCP in the transportation sector is still in its early stages. However, technological advances and the proliferation of competitors have intensified competition in the transport sector, forcing providers to prioritize customer retention to remain competitive. In this context, accurately predicting customer churn becomes crucial, especially in markets in which there is no contractual relationship with the customer (Ahn et al., 2020; Lee et al., 2011; Miguéis et al., 2013). Traditional churn prediction models often focus on maximizing overall accuracy, which may not directly translate to improved business outcomes. However, by incorporating profit-driven approaches, transport providers can make more informed decisions about customer retention strategies. These approaches prioritize the identification and retention of high-value customers, maximizing revenue.

One example of an industry which operates in a non-contractual setting is the taxi industry, where a single search in a mobile application, allows a customer to get an estimate of the price to pay for a ride and decide whether he/she opts for a taxi or other transport mode.

Contrary to other alternative modes of transportation, in some countries, taxi fares are regulated, meaning that taxi providers are not allowed to offer direct discounts or rewards to customers. Under these circumstances, the retention of customers and attraction of new ones is an even more challenging task faced by taxi providers, who need to adopt differentiated strategies to continuously stimulate customers' interest.

Although still scarcely explored in the transport context, customer attrition prediction was the main topic of a few works addressing the airline industry and road shared mobility. For instance, Park et al. (2022) applied a set of classification machine learning techniques to evaluate the satisfaction and churning risk of airline customers using indicators related to physical (e.g. cabin conditions) and social environment (e.g. amount of travelers) of the airlines along with brand experience (e.g. kindness of cabin crew) and loyalty (e.g. willingness to flight again with that company). The authors analysed data gathered from a survey, having concluded that, the prediction accuracy of most models is increased when social and physical aspects are taken simultaneously. In Hu et al. (2021), the impact of customer demographics, stations' locations and time-varying personalized economic rewards in car-sharing customer attrition propensity was assessed. Applying a survival model, the authors estimated customer churn likelihood over time, having displayed it under the form of a survival curve. In this study, it was demonstrated that the persuasive effect of economic rewards is much more effective over a short initial period. Moreover, elderly and male customers were identified as the most loyal ones, and the authors suggest that car-sharing locations should be situated away from transit hubs.

While substantial research has been conducted to predict customer churn across various domains, this area remains relatively unexplored within the transportation sector. This study aims to contribute to a better understanding of customer retention in the transportation by exploring CCP within this domain. Furthermore, we aim to bring the concept of profit-driven CCP, which has been successfully applied in other sectors, to the taxi industry by developing models that directly optimize for profitability. Utilizing a comprehensive dataset encompassing service records from an entire fleet over a 52-month period, we investigate CCP within taxi services. Furthermore, since the data contains a unique identifier for each customer, we can apply a set of machine learning techniques to characterize the recent travel behavior of individual customers and infer which customers are at higher risk of defecting. Therefore, in this work, the one-to-one relationship between the service provider company and the customer is explored in the context of the taxi industry, aiming to support companies in developing more effective retention actions.

3. Methodology

The aim of this study is to detect potential churners early by analysing the past travel behavior of each customer. Additionally, we

want to concentrate efforts on customers with a certain level of engagement with the company. Therefore, and following the opinion of experts, we decide to narrow down the customer selection criteria, focusing on customers who have recorded a minimum of 2 services per month. Customer data is grouped into monthly intervals. Subsequently, a three-month observation window is employed to characterize customer travel behavior and predict their churn probability in the subsequent month (hereafter referred to as the target month). The selection of a three-month characterization period was guided by prior research, notably the work of Migueis et al. (2012), and validated through consultation with domain experts to ensure its relevance within the specific context of the taxi industry. To accomplish the goal of the study, we employ different machine learning methods, and evaluate the effectiveness of these methods, based on a set of metrics. For each method we run two models: one considering the complete set of predictive variables and the other in which the best combination of predictive variables is found through a greedy feature forward selection approach. To evaluate the significance of the computed models, the performance attained for each model is compared with the performance of a baseline model. We define the baseline model as the one that assumes all observations as churners and use it to evaluate the effectiveness of other models and to select the best classifier. In the remainder of this section, we detail the criterion employed to infer a churning behavior, propose the predictive variables to integrate the classification models, describe the evaluation procedure and metrics used to measure the performance of the models, and we also provide a brief description of the machine learning classification techniques applied.

3.1. Churning behavior

In a non-contractual setting, the occurrence of a churn event is not precisely defined. Then, to construct a binary classification model, it is required to delineate a criterion to distinguish churners from non-churners, based on which the target variable of the model is set. Accordingly, for each customer, after gathering the monthly services data, we calculate the average number of services in the first three months of contact with the company. Then, a customer who, after the first three months, registers in all subsequent months a decrease in the number of services of more than 40% compared to the recorded average in the initial three months is qualified as a churner. The threshold value of 40% for customer churn was established based on a combination of prior research and experts' opinion. This value aligns with the work of Migueis et al. (2012), who identified a similar threshold in their study. Furthermore, we consulted with domain experts to validate this threshold, ensuring its relevance within the specific context of the taxi industry. However, as stated by Clemente-Ciscar et al. (2014) the definition of churn in non-contractual environments is not standardized and may be subject to continuous refinement. After the first event as churner, the customer is pulled out from the predictive models, ensuring that no predictions are made concerning customers who have already defected.

3.2. Predictive variables

This work aims to achieve the objective of distinguishing between churners and non-churners among customers, based on their past travel behavior. For this purpose, we suggest to analyze the travel data of customers using a set of variables that measure the recency and frequency of taxi service usage. Recency refers to how recently a customer has used a service and measures the time elapsed since the customer's last interaction (Chai et al., 2023; Chen et al., 2022). Previous studies (e.g. Buckinx and Van den Poel (2005)) conclude that customers who have used a service more recently (lower recency values) are likely to have a stronger connection with the service provider, and then are less prone to churn. Therefore, in the context of customer churn analysis, recency can indicate the potential for churn. On its side, frequency measures how often a customer uses a service within a given time period (Mosa et al., 2023). Higher frequency suggests a stronger engagement and reliance on the service. In the context of customer churn analysis, lower frequency values suggest a higher churn risk, as customers with declining usage may indicate decreasing commitment (Bolton et al., 2004; Miguéis et al., 2012). In summary, recency and frequency provide insights into a customer's recent activity, changes in behavior and engagement level.

Additionally, we propose the inclusion of predictive variables associated with the customers' satisfaction level, such as the waiting time experienced by customers or the ratio between the waiting time and the effective time duration of the service. Note that for waiting time we mean the period that elapses between the moment when the customer requires the service and the moment when the taxi picks up the customer.

To capture trends and changes in behavior, we also recommend to integrate predictive variables expressing the variation of the other variables. In an effort to highlight the more recent behavioral changes of the customers, we suggest incorporating the percentage of variation observed during the last two months preceding the target month for specific dimensions. As an example, for the number of services, we propose the following variables: (i) the total number of services recorded in each of the 3 months of the characterization period; (ii) a categorical variable indicating the trend in the number of services over the 3-month characterization period (e.g. whether there is an increase in the number of services over the past three months or an increase followed by a decrease in this number); (iii) percentage variation in the number of services recorded in the last two months of the characterization period, computed as the ratio of the difference between the number of services performed in the last two months and the number of services in the second last month of the characterization period.

We suggest that depending on the predictive variable, the value considered may correspond to its minimal, average or maximum value. The reasoning behind the chosen value should be related to the measure considered most likely to result in churn behavior.

Furthermore, we consider that more general variables, such as the day of the week or the period of the day on which the service was performed, should also be part of the predictive variables set.

Lastly, we suggest integrating spatial data describing the taxi path in the models, namely the pick-up and drop-off locations and the distance traveled.

By incorporating these variables into churn prediction models, we believe that businesses can identify potential churners and take proactive measures to retain them.

Our churn prediction methodology, developed for the taxi industry, exhibit similarities with customer retention strategies in several sectors, namely retail. Both sectors operate in non-contractual environments with low switching costs for customers. Our approach, leveraging recency and frequency measures, can be readily adapted to retail. This demonstrates the broader applicability of our methodology across industries where customer loyalty is a key business objective.

3.3. Model selection and evaluation

In this study, the predictive power of binary classification models is measured focusing on the EMPC, lift and F1-score. The latter is adopted to compare the performance of the developed models with a baseline model for which EMPC calculation was not feasible (refer to Section 5.1). F1-score values range from 0 to 1, with values closer to 1 revealing higher discriminatory power Sitarz (2023). To better evaluate the benefits of retaining customers and the costs associated with retention campaigns, Verbraken et al. (2013) introduced a cost-benefit analysis framework tailored for customer churn. In this framework, the average classification profit function is expressed as:

$$PC(t; c, \mathrm{CLV}, \delta, \phi) = \left\{ [c \cdot (1-\delta) - \phi] \cdot \pi_1 \cdot F_1 - (\delta + \phi) \cdot \pi_0 \cdot F_0 \right\} \cdot \mathrm{CLV}. \quad (1)$$

Here, t represents the classification threshold, CLV denotes the customer lifetime value per retained customer, and c is the probability of successfully converting a churner into a non-churner. The parameters $delta$ and ϕ are calculated as $delta = d/CLV$ and $\phi = f/CLV$, where d is the cost of successfully retaining a churner and f is the fixed cost of contacting a customer. Building on this profit function, the EMPC is extended as follows:

$$\text{EMPC} = \int_c PC(T(c); c, \text{CLV}, \delta, \phi) \cdot h(c)\, dc \tag{2}$$

In Equation (2), T represents the optimal classification threshold, and $h(c)$ is the probability density function of c, which follows a beta distribution. The probability density function $h(c)$ is defined as:

$$h(c) = h(c; \alpha, \beta) = \frac{c^{\alpha-1}(1-c)^{\beta-1}}{\int_0^1 u^{\alpha-1}(1-u)^{\beta-1}\, du}. \tag{3}$$

In line with Maldonado et al. (2020), Hoeppner et al. (2020), Jiang et al. (2024) and Liu et al. (2025), for the EMPC calculation, we have adopted the recommended default values for customer retention campaigns, suggested by Verbraken et al. (2013). More precisely, CLV was set at 200€, d was set at 10€, and f was set at 1€. Furthermore, the probability parameter c in the EMPC was assumed to follow a beta distribution with shape parameters defines as $\alpha = 6$ and $\beta = 14$.

In addition to EMPC, we compute lift, an evaluation metric commonly used in marketing campaigns to evaluate the effectiveness of targeting specific customer segments corresponding to the top x%. To calculate lift value, the response rate of the targeted group is divided by the overall response rate (Lee et al., 2011). In this study, we use lift to focus on the customer segment with the greatest likelihood of churning. Thus, in this context, lift quantifies how much the model under evaluation performs better in predicting churners in top x% than a random selection. A lift value higher than 1 demonstrates the effectiveness of the model to target high-risk churn customers to improve customer retention. By targeting specific customer segments with a higher lift, retention campaigns can generate better results and optimize their return on investment. Lift values obtained when evaluating the model considering only the 5% to 30% of customers predicted most likely to churn are presented.

In this research work, we propose a temporal division of data for hyperparameter tuning and model evaluation purposes. The objective of this procedure is to identify the predictors and hyperparameter configurations that lead to higher model performance for each machine learning technique. Afterwards, the performance of the best model is evaluated when applied to unseen data. To accomplish this, the initial dataset is split is training, validation and testing sets.

The methodology employed in this study and described in previous paragraphs is illustrated in Fig. 1. After grouping the data for each customer into monthly periods, we examine the data within 3 month periods to define the travel behavior of the customer. Accordingly, we identify this 3 month period as the characterization period. Then, we forecast the likelihood of the customer churning in the subsequent month, ie, in the target month. This procedure is repeated throughout the full dataset, shifting forward the 3 month period by one month at a time. For example, one analysis will use the data from January, February and March to predict the churning behavior of a costumer in April, while the subsequent analysis will use the data for the months of February, March and April to predict the same for the month of May.

The optimal combination of hyperparameters and predictors is identified by training each combination on the training set and evaluating its performance on a hold-out validation set. Regarding predictor selection, a greedy forward feature selection approach is utilized. Initially, each predictor is individually assessed in relation to the dependent variable. Subsequently, a hyperparameter optimization process is executed, evaluating the performance of the resulting model on the validation dataset. Hyperparameter tuning is performed using a brute-force grid search. The combination of predictive variables and hyperparameters that maximize the EMPC value is selected. The process continues by combining the individual predictive variable with the remaining ones, iterating until no further improvement in the EMPC value is observed. A non-improvement in EMPC is indicative that additional predictive variables do not enhance the predictions. The optimal combination of predictive variables and hyperparameter settings that yield the best-performing model is used to predict churners. In line with Maldonado et al. (2020) and Jiang et al. (2024), the model that achieved the highest EMPC value was considered the best performer. Such model is then trained on the training-validation set and evaluated on the testing set to ensure its effectiveness. To support the preceding description and provide a clearer understanding of the model development process, the pseudo-code of the algorithm's structure is presented in Fig. 2. By following this procedure, using separate training, validation, and testing sets, predictive models can be evaluated, hyperparameters can be fine-tuned, and the best-performing model can be selected for deployment, while avoiding overfitting and increasing the models' robustness.

Note that, feature selection could be performed by leveraging the inherent feature importance estimates provided by Random Forest (RF). However, while RF is commonly used for both prediction and feature selection, its feature importance estimates can be deprecated when dealing with a high proportion of irrelevant variables (Olaniran and Alzahrani, 2023). Moreover, the presence of irrelevant features can negatively impact model performance, as demonstrated by the work of Kim and Kang (2019). To maintain consistency and avoid algorithm-specific biases, we employed a uniform feature selection process across all models.

3.4. Machine learning techniques and hyperparameters tuning

As previously mentioned, we investigate the utilization of different machine learning techniques to distinguish churners from non-churners: Light Gradient Boosting Machine (LightGBM), Random Forest (RF) (Breiman, 2001) and Artificial Neural Networks (ANN) (McCulloch and Pitts, 1943). Furthermore, to enhance the efficiency of the predictive models, as mentioned before, we carry out a customized hyperparameter tuning process. To achieve this, a brute-force grid search method is used to explore various parameter values for each technique. The choice of LightGBM, RF and ANN for predicting customer churn in grounded on their widespread use in the literature, where these machine learning techniques have consistently demonstrated have good performance when handling complex churn prediction tasks. Additionally, an ensemble stacking model is developed, better aligning this work with current research emphasizing the enhanced predictive performance of combined models in such applications (Jiang et al., 2024). In the following paragraphs, a brief description of each employed technique is provided, while the hyperparameters selected for tuning, and their corresponding tested values are detailed in Appendix A.

Light Gradient Boosting Machine

LightGBM is a machine learning algorithm that belongs to the family of gradient boosting methods and, as such, it is based on decision trees. It utilizes a gradient boosting framework, where multiple weak prediction models are sequentially trained to correct the errors made by the previous models. However, what sets LightGBM apart is its unique approach to constructing decision trees and handling the training process. One key feature of LightGBM is the gradient-based leaf-wise tree growth strategy. Instead of growing trees level by level, LightGBM grows trees in a leaf-wise fashion. It selects the leaf with the maximum loss reduction to expand at each iteration, resulting in a more effective use of the available data. Additionally, LightGBM implements a technique called histogram-based binning, which discretizes continuous features into discrete bins. This enables faster and memory-efficient training by reducing the number of splits to be evaluated during the tree building process. Furthermore, LightGBM employs a technique called Gradient-based One-Side Sampling (GOSS) to reduce the computational cost of

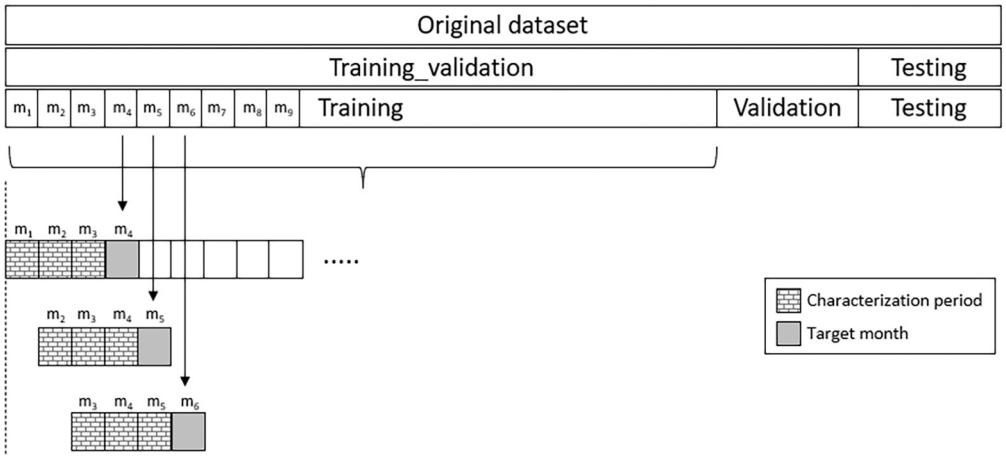


Fig. 1. Schematic representation of the methodology followed.

```
Partitioning of the dataset into training, validation and testing sets
WHILE the current EMPC is higher than the last EMPC DO
  FOR each combination of variables (current set of predictors and a new variable)
    FOR each possible value of parameter_1
      FOR each possible value of parameter_n
        Train the predictive model on the training set
        Compute predictions on the validation set
        Calculate the EMPC between predictions and the values of the validation set
      END FOR
    END FOR
  END FOR
Store the highest value of EMPC as the current EMPC
END WHILE
Store the combination of predictive variables and parameters that result on the best performing
model
Train the predictive model on the training_validation set
Compute predictions on the testing set
Calculate the EMPC between predictions and the values of the testing set
Calculate the remaining performance metrics of the model
```

Fig. 2. The pseudo-code of the algorithm' structure.

training. GOSS keeps the instances with large gradients while randomly sampling the instances with small gradients, thus providing a good balance between reducing overfitting and maintaining accuracy. Overall, LightGBM is known for its high efficiency, scalability, and accuracy in handling large-scale datasets (Zhu et al., 2023).

Random Forest

RF has its roots in decision trees and is designed to address the limitations of this concept, aiming to enhance their accuracy and overcome issues such as their sensitivity to minor data variations. It involves building an ensemble of trees and calculating the average prediction from the final nodes of each tree. By doing so, RF mitigates the lack of robustness observed in individual decision trees. Additionally, this approach randomly selects a subset of independent variables to grow each tree (Breiman, 2001).

Artificial Neural Networks

ANN are a collection of computational tools built using interconnected processing units known as neurons. Each connection between neurons is assigned a weight. These networks are structured in layers, with each neuron receiving inputs from the neurons in the previous layer and producing a nonlinear weighted sum as output to the connected neurons in the subsequent layer. These weighted connections enable the model to learn the relationships between variables. A typical feed-forward neural network comprises an input layer, a hidden layer, and an output layer. Activation functions are applied to the input values of neurons in the hidden and output layers (Han et al., 2011). Numerous neural network architectures and learning algorithms have been developed, with backpropagation emerging as the most prominent training method due to its demonstrated effectiveness. Given its widespread use and successful track record, the backpropagation algorithm was also employed in this study.

Stacking ensemble model

Due to the inherent complexity of CCP, individual predictive models are not always effective to achieve high accuracy (Idris et al., 2013). Ensemble learning, which combines predictions from multiple models, has emerged as a promising solution. Among the three main types of ensemble methods (bagging, boosting, and stacking), stacking has shown particular promise in improving the accuracy of CCP (Rokach, 2010). Stacking classifiers distinguish themselves through a unique approach that potentially yields higher accuracy. Unlike bagging and boosting, which typically combine similar types of base models, stacking can leverage both homogeneous and heterogeneous ensembles. In a stacking model, all ensemble members are trained on the original data. Subsequently, their predictions on a separate validation set (or a portion of the training data) are treated as new input features. A higher-level "meta-

classifier" is then trained on these derived features to generate the final prediction. This flexibility, particularly when combining diverse base models, allows stacking to potentially achieve superior performance compared to bagging or boosting (Jiang et al., 2024; Liu et al., 2025). In this work, feature selection and a brute-force grid search are utilized to guide the training process and optimize the hyperparameters of a diverse set of profit-driven classifiers that are used as ensemble members. Subsequently, a linear classifier is employed as a meta-learner to combine the outputs of the ensemble members, aiming to construct a more accurate and robust final model.

3.5. Model interpretability

While ensemble models offer powerful prediction capabilities, they can be opaque in their decision-making process. To address this interpretability challenge, we leverage Shapley Additive Explanations (SHAP) values, first introduced by Lundberg and Lee (2017). SHAP values act like a feature attribution method, quantifying the individual contribution of each feature to a specific prediction. They achieve this by simulating feature removal and measuring the resulting change in the model's output (Lundberg et al., 2020). This allows us to understand not only which features are important globally (by averaging SHAP values across all data points) but also how each feature influences individual predictions (acting as a local interpretation method).

4. Case study and data

In this research work, we use the data of the services carried out by the fleet of taxis operating in the Portuguese city of Coimbra as a case study. All taxis of this fleet belong to the same cooperative, meaning they are all connected to the same dispatching centre. Furthermore, they are equipped with an onboard electronic device that continuously gathers and stores data. The company responsible for developing and installing these devices provided us with the data collected between October 2014 and December 2018 (totaling 51 full months of data).[1] More specifically, we explore the data of services requested via phone call, mobile application or through the website of the cooperative to which the taxis belong. Services departed from taxi stands or directly hailed by customers to taxis cruising vacantly are excluded, as we have no access to a customer identifier in these cases.

The original dataset provided has more than 1.5 million observations, referring to the same number of taxi services, described by means of 6 variables. The set of variables includes the identifiers for service and customer, the means used to request the service and timestamps of the following instant times: service creation, customer pick-up and customer drop-off. It is worth noting that cases of missing or inconsistent data were not identified, and as a result, we proceeded to the features estimation without conducting any other king of preprocessing procedures. To provide the reader with a more comprehensive understanding of the taxi business in Coimbra, in the subsequent paragraphs, we present a collection of descriptive statistics pertaining to the taxi operations within the period under analysis. A detailed analysis of the data reveals a consistent increase in the number of services performed between the years 2015 and 2018, with growth rates ranging from 19.5% in 2015 to 27.9% in 2018. Besides that, the services analysed were requested by a total of 155,480 distinct customers, 43.2% of whom requested only 1 service over the entire period and 84.8% of customers recorded up to 10 services in the same period. Furthermore, only 145 customers requested at least 1 service per month. Based on these findings, we can infer that in Coimbra, the majority of taxi users utilize this mean of transport on an irregular or non-regular basis. This fact also highlights the importance of strengthening connections with regular customers, reducing the business's dependency on occasional customers. Examining the temporal distribution of services, it is clear that August records the lowest percentage of services, accounting for only 5.8%. In contrast, October stands out with a significantly higher percentage, representing more than 11.1% of the total services. We believe that the fact that August is traditionally the holiday month for many residents in Portugal, who travel outside their area of residence, may significantly contribute to this difference. Furthermore, the demand for services is higher in the morning (between 6am and 12am) and lower in the night (between 9pm and 5am), varying from 33.9% to 15.6%. In addition, most of the services are requested in working days (78.5%), against only 21.5% on weekends. Regarding the preferred mean used for requesting a taxi, the majority of customers (98.9%) opt for making a phone call, while only 0.9% use the taxi-hailing APP. A very small percentage (0.1%) use the cooperative's website for their requests. On average, customers experience a waiting time of less than 4 minutes for a taxi, and the duration of a typical taxi service is approximately 13 minutes. Fig. 3 shows the number of services requested and the passenger waiting time over the period under analysis. Analysing the figure we can observe that while the decrease in the number of services in August is evident, there is no significant variation is observed in the average waiting time. Nonetheless, the plot still shows some correlation between the number of services and the average waiting times, as the months with a higher number of services appear to result in a slight increase in the waiting times, while the reverse is also true.

After a pre-processing step, the final dataset used in the machine learning models is composed by a total of 47 predictors (excluding service and customer identifiers) that are presented in Table B.5. Aligned with what is proposed in Section 3.2, the final set of predictors contains variables expressing the recency and frequency of taxi service use, together with predictors that capture customers' satisfaction level, trends and behavior shifts and also more general variables. However, we were not provided with any spatial data of the services recorded and, therefore the models developed do not account for this data.

As mentioned in Section 3, we group the data of each customer in 1 month periods and restrict our analysis to customers with at least 2 services per month, focusing on those who consistently use the services of the company. This procedure leads to a final dataset of 29,444 observations, with each observation corresponding to a unique combination of customer/month. In case of the categorical predictors such as the period of the day or the day of the week, when grouping the data, we determine the most frequent value recorded in each month for each customer. Then, the target variable is determined, according to the criteria described in Section 3.1, resulting in a final distribution of 61.3% churners and 38.7% of non-churners. To implement the hold-out method for hyperparameter tuning and model evaluation purpose, we use the data of the first 45 months for training, the data of the following 3 months for validation and the last 3 months for testing. It is worth mentioning that the predictive variables considered fall within the bounds of the data made available.

5. Results and discussion

In this section, we present and discuss the results gathered in this study. Firstly, we present the outcomes of the models run when considering the complete set of predictors (Section 5.1). Then, we shift our focus to the performance of the models developed using a greedy forward feature selection approach (Section 5.2). Additionally, we devote attention to the interpretability of the predictive results (Section 5.3). Finally, we point out managerial implications and suggest retention actions that companies can follow to benefit from the findings of this study (Section 5.4).

[1] It is worth mentioning that in this period no ride-hailing platform was operating in the city.

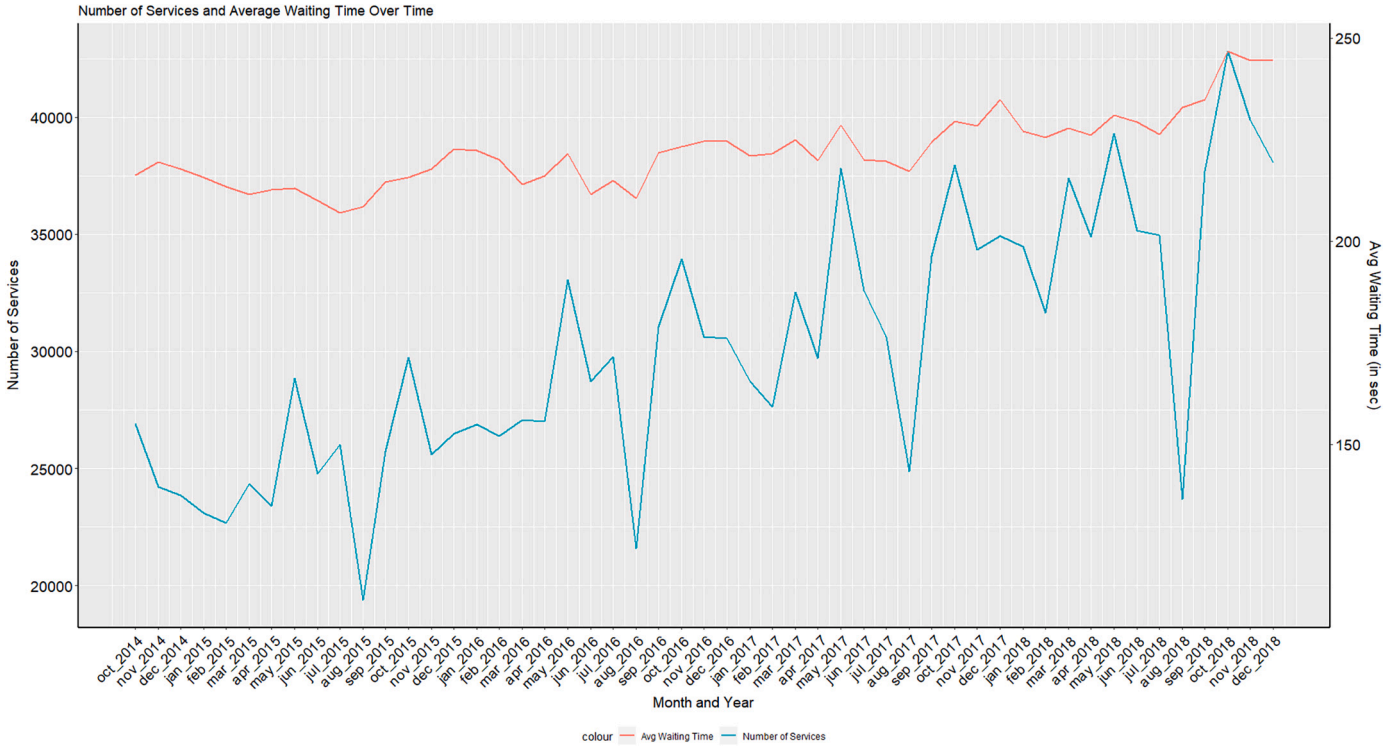


Fig. 3. Number of services and Passenger waiting time over time.

Table 1
Classification models performance when considering the complete set of predictors.

Technique	Evaluation metrics		
	F1	AUC	EMPC
LightGBM	0.6793	0.7410	37.5376
RF	0.6729	0.7329	36.2818
ANN	0.5876	0.5911	34.2818
Ensemble model	0.6335	0.7294	32.9483
Baseline model	0.5485		

5.1. Modeling with the complete set of predictive variables

For each technique used to predict the churning behavior of customers, a model is run, considering the complete set of predictors, and respective hyperparameters are fine-tuned using an exhaustive grid search approach, covering the range of values presented in Table A.4. Unlike the other models, the Stacking Ensemble model does not undergo hyperparameter tuning. This is because it leverages the predictions of LightGBM, RF, and ANN as its input features. Table 1 provides a summary of the results attained for each technique and for the stacking ensemble model, together with the performance obtained by a baseline model, based on the various evaluation metrics employed. The hyperparameter values that allow us to obtain the best level of performance of each technique are presented in Appendix C (Table C.6).

To validate the effectiveness of the models, we compared their performance to a baseline model, which assumes all customers as churners. Since EMPC was unavailable for the baseline model, F1-scores were used for comparison, revealing that the baseline model attained a significantly lower F1-score. This confirms that all applied machine learning models substantially outperform a naive prediction approach, demonstrating a higher capability to predict churners. This finding reinforces the value of the developed models as tools to aid a company's decision-making process. The results also indicate that, regardless of the evaluation metric chosen, LightGBM consistently exhibits the best performance, making it the top-performing model. Indeed, LightGBM achieved the highest EMPC (37.54), indicating it is the most profitable model for predicting customer churn. Also, its higher AUC value (0.74) confirms its superior ability to effectively distinguish between churners and non-churners. Another finding is that, from a profitability perspective, the stacking ensemble model did not demonstrate any advantage over individual models, obtaining the lowest EMPC value, which suggests that ensembling did not lead to substantial profit gains in this case. Overall, LightGBM emerges as the most profitable model, followed closely by RF (3.36%), while ANN (9.50%) and the ensemble model (13.93%) demonstrate lower financial viability. Within parenthesis are reported the percent improvement in EMPC value of the LightGBM algorithm over the others.

As mentioned in Section 3.3, the predictive power of the models when targeting specific proportions of higher risk customers is also measured by determining the respective lift value. In Fig. 4 is graphically represented the performance of the models when compared to a random model.

It is worth noting that the RF model correctly identifies approximately 90% more churners than a random model when targeting the top 5% of customers with the highest defection propensity. At the same targeting level, the ANN and Ensemble models correctly classify around 50% more churners than a random model, while LightGBM, though performing less strongly in this specific context, still identifies 30% more churners than a random model. As expected, increasing the proportion of targeted customers (and thus including customers with progressively lower churn probabilities) leads to a decrease in lift values across all models. This trend is particularly evident for the RF model. Conversely, while the Ensemble model's lift curve exhibits an unusual upward trend at lower cut-off values, it demonstrates greater stability compared to the other models. The ANN model also displays consistent lift values across all target percentiles.

5.2. Modeling with best performing combination of variables

Resorting to a greedy feature forward selection approach, we determine the best combination of predictive variables for each technique

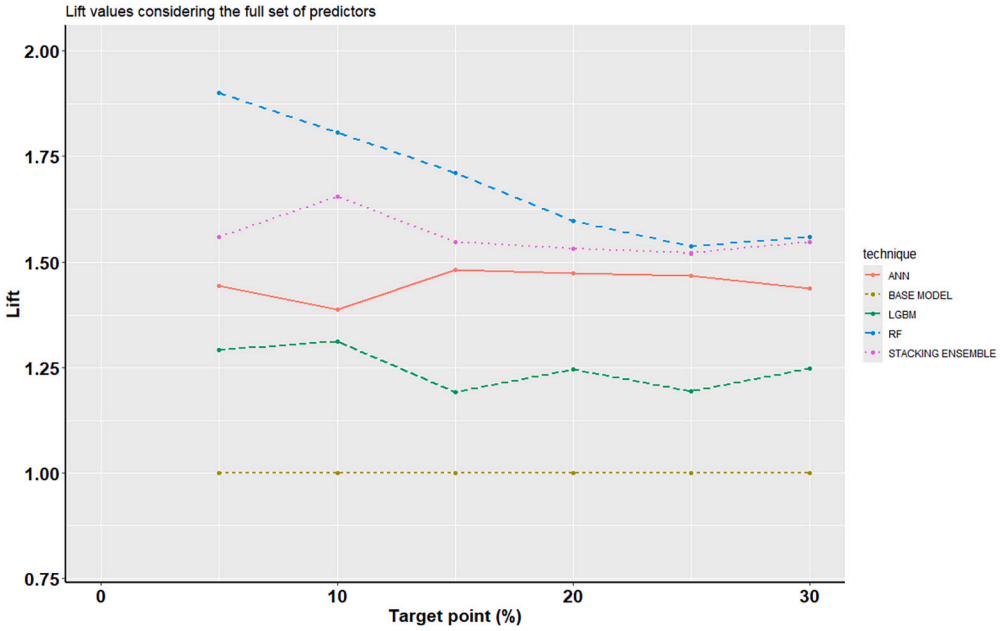


Fig. 4. Lift values trends of predictive models compared to a baseline model, when running models with the full set of predictors.

Table 2
Best performing combination of variables for each technique.

Technique	Combination of variables
LightGBM	n_services_prev_homol_month + holiday_month + usual_day_type_m1 + usual_service_subtype_m1 + usual_service_subtype_m2 + usual_service_subtype_m3
RF	n_services_prev_homol_month + holiday_month + avg_delta_time_m2 + avg_delta_time_m3
ANN	avg_wait_duration_m2 + avg_delta_time_m1

employed. This approach entails that the inclusion of an additional predictive variable only occurs when it contributes to improve the model's performance. It is an iterative process that finishes when the addition of any other variable is not reflected in an increase in EMPC value. Once again, for each combination of variables, most favorable hyperparameter values are found by way of a brute-force grid search. Consequently, the selection of variables that yield the best results varies for each technique, reflecting the unique characteristics of each algorithm. Table 2 presents the combinations of variables derived from each technique, while Table 3 summarizes the final evaluation metrics for each technique. The best performing hyperparameters gathered for each model are presented in Appendix C.

The analysis of Table 2 highlights the consistent inclusion of the number of services in the same month of the previous year and the information of being or not a traditional holiday month for both LightGBM and RF models. This finding seems to suggest that: (1) the comparison between the travel behavior of customers in the same month of the previous year and their most recent travel patterns could indicate a notable shift in their behavior; (2) traditionally holiday months for the general population, demonstrate to have a significant impact on the travel behavior of taxi users. Furthermore, our analysis reveals that the customer's preferred method for requesting services, the average duration of taxi services, and the proportion of waiting time relative to service duration are relevant factors in predicting the likelihood of encountering customer attrition. However, it is surprising to note that the inclusion of additional variables, such as services variation in the last 2 weeks and time elapsed between the last service in the previous month and the first day of the target month, which are intended to capture the more recent customer behavior, does not seem to enhance the performance of the models. This is evident from the fact that these variables are not present in any of the best performing variable combinations.

Table 3
Classification models performance when applying feature selection.

Technique	Evaluation metrics		
	F1	AUC	EMPC
LightGBM	0.5937	0.6363	42.8046
RF	0.5896	0.6275	40.7931
ANN	0.4829	0.5152	35.3218
Ensemble model	0.5971	0.6977	41.6379
Baseline model	0.5485		

A comparison between Table 1 and Table 3 reveals that optimized models with selected predictors outperform their full-variable counterparts in terms of profitability. This reinforces the advantage of a feature selection process to enhance financial outcomes. LightGBM remains the best choice for maximizing profit, while RF and the Ensemble model provide strong alternatives. However, ANN continues to exhibit lower performance, making it a less favorable option. In summary, the findings suggest that feature selection enhances model profitability, with LightGBM maintaining its dominance across both settings. The increase in EMPC for RF and the ensemble model further supports the conclusion that a refined selection of variables leads to better financial performance in churn prediction models. On the other hand, the feature selection process seems to lower model capacity in effectively distinguish between churners and non-churners, as all models present a lower AUC value when trained with the best performing predictors. Overall, all predictive

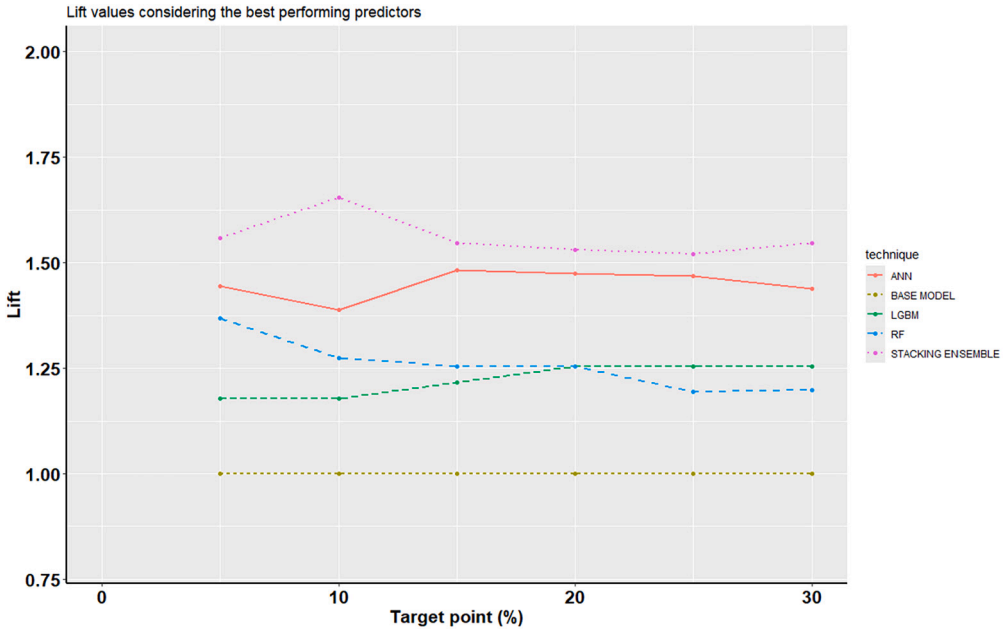


Fig. 5. Lift values trends of predictive models compared to a baseline model, when running models with best-performing predictors for each technique.

models, using the best-performing predictors, outperformed the baseline model in churn prediction, with the exception of the ANN model.

Similarly to models with all predictive variables, the accuracy of predictions gathered when considering the best performing variables are determined for varied targeting proportions of customers. Fig. 5 shows the lift curves for the models run with the best combination of variables. Regardless of the selection point, the lift values for all employed techniques are notably above those attained with a random selection. Furthermore, unlike the models trained using all variables, the ANN model classifies a higher proportion of churners than both the LightGBM and RF models across all cut-off points. Specifically, when targeting the top 5% of customers at highest risk of churn, the ANN model correctly identifies 44% more churners than a random model, while the RF model identifies 36% more. Analysis of Fig. 5 reveals that the RF model exhibits significantly higher predictive power than LightGBM for cut-off values up to 20%. However, this advantage diminishes when targeting a broader customer group, with LightGBM surpassing the predictive capacity of RF. Comparing Fig. 4 and Fig. 5, the most significant difference is in the lift values achieved by the RF model. The RF model shows a notable decrease in its ability to correctly identify churners, even being surpassed by the ANN and LightGBM models for target percentiles above 20%. This reduction in performance can be attributed to the fact that the best-performing variables were selected solely based on EMPC. In contrast, using this more restricted set of predictive variables does not significantly impact the lift values of the ANN and LightGBM models. Notwithstanding the above, we still consider that although the predictive capability of the models is lowered, this approach is still advantageous, since it reduces computational costs and has a lower level of complexity.

It is worth noting that the AUC values attained by the predictive models show a relatively low overall performance. However, the developed models can be deemed as effective models for predicting customer churn, particularly for customers with a high propensity to churn. In practice, retention campaigns often prioritize these customers, making these models valuable for optimizing marketing efforts. The proposed models exhibit strong performance, surpassing simpler alternatives. Additionally, by pinpointing key churn indicators, businesses can collect actionable insights to refine their retention strategies, ultimately boosting profitability and minimizing resource costs.

5.3. Model interpretability

Beyond predictive accuracy, understanding how a model arrives at its predictions is essential. SHAP values offer a powerful method for interpreting model behavior by revealing the influence of each feature on the final prediction, where the magnitude of the SHAP value indicates the strength of that feature's impact. A negative SHAP value signifies that a particular feature decreases the likelihood of the predicted outcome, while a positive value suggests that the feature increases the likelihood. Fig. 6 presents the SHAP values for the LightGBM model, which achieved the highest EMPC value using the set of best-performing predictors. In this figure features are presented in descending order of the mean absolute SHAP value across all instances.

In few words, if a feature's SHAP values exhibit a predominantly positive association with customer churn (i.e., more blue dots with negative SHAP values and more red dots with positive SHAP values), then higher values for that feature generally indicate a higher likelihood of churn. Conversely, if the association is predominantly negative (more red dots with negative SHAP values and more blue dots with positive SHAP values), higher feature values are associated with a lower likelihood of churn. Fig. 6 demonstrates a strong correlation between the predictive output and the customer's preferred method for requesting taxi services, namely for customers who favor the website and the mobile application. This finding can be attributed to the fact that these customers may be more exposed to competing services through digital channels and may be more likely to switch to alternative transportation options. Furthermore, the model assigns higher importance to the summer months (July, August, and September, where '*holiday_month*' = 1), as indicated by positive SHAP values. Additionally, weekends also contribute positively to model predictions. These findings may suggest that seasonal variations and weekend travel patterns significantly impact customer churn behavior. These findings underscore the observation that a significant proportion of taxi users exhibit irregular or infrequent usage patterns, stated in Section 4. Furthermore, customers who utilized the taxi service more frequently in the preceding year, as indicated by a higher number of services in the homologous month of the previous year, exhibit a greater propensity for churn. This finding underscores the critical importance of maintaining consistent engagement with frequent customers to foster long-term loyalty. Summing up, machine learning models, when combined with SHAP value analysis, provide powerful interpretability,

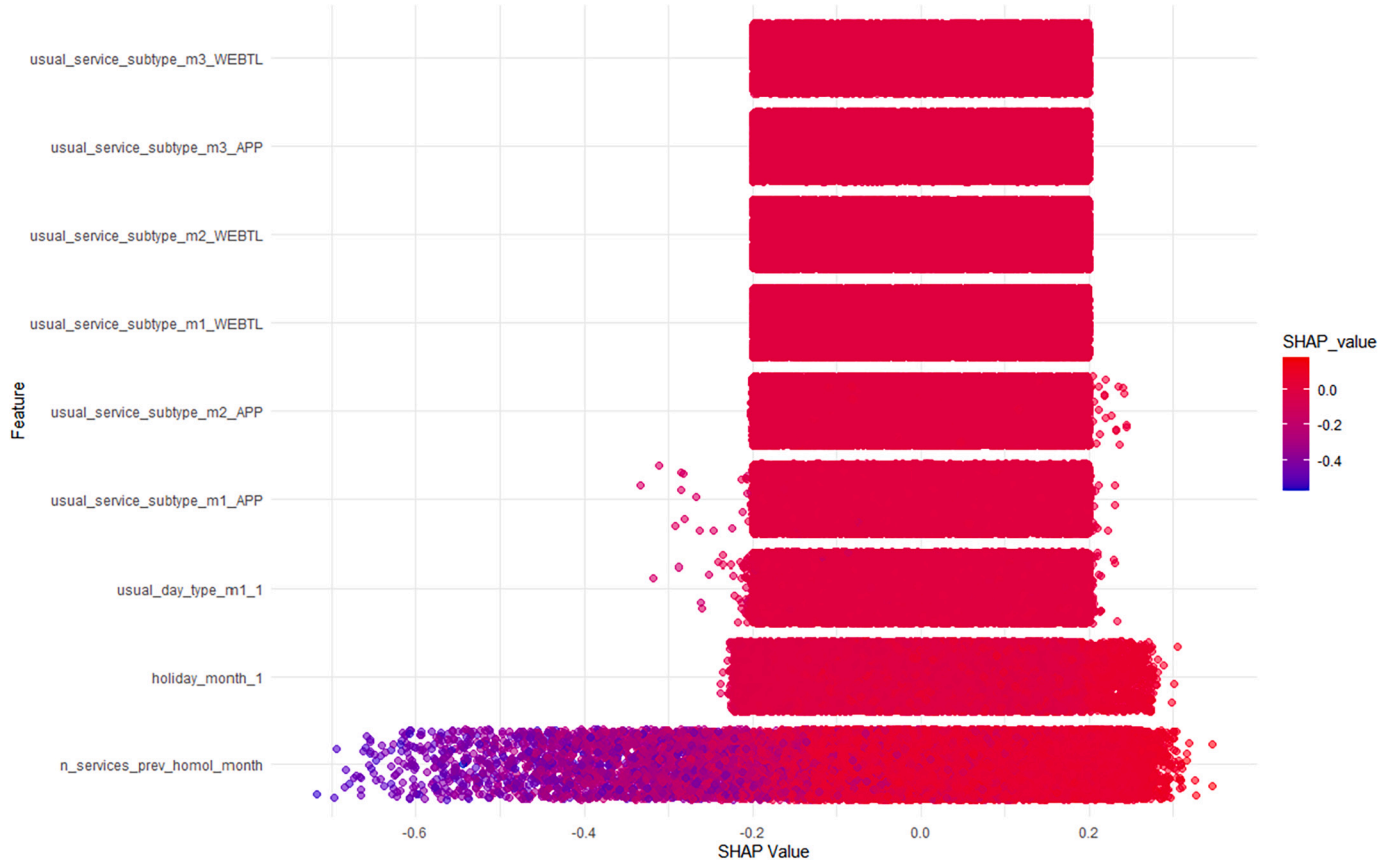


Fig. 6. SHAP values for the LightGBM model with best performing combination of variables.

with SHAP values effectively elucidating the strength and direction of each feature's influence on the model's predictions.

5.4. Managerial implications and retention actions

Unlike other industries such as banking or telecommunications, where churn often involves contract termination and switching costs, customer attrition in the taxi sector is non-contractual and could be volatile. This makes churn prediction particularly challenging, as customers may temporarily disengage rather than fully defect. This study demonstrates that advanced machine learning, particularly LightGBM, significantly enhances churn prediction in the taxi industry. Profit-driven models and refined feature selection are crucial for maximizing predictive accuracy and profitability. Our findings underscore the necessity of using advanced machine learning models over traditional approaches, as they allow for a more nuanced understanding of customer behavior and enable dynamic, data-driven decision-making. By integrating predictive modeling, taxi companies can move beyond reactive churn mitigation and adopt proactive engagement strategies that promote customer loyalty. The insights from this study provide a foundation for more effective, targeted retention efforts, ensuring that marketing investments yield the highest possible returns in a competitive and rapidly evolving mobility landscape. A key managerial takeaway is the ability to segment customers based on the most influential churn indicators. Our results show that seasonal usage patterns and booking methods play a crucial role in churn behavior. In particular, customers who primarily use digital booking channels (app or website) face a higher risk of churn, likely due to increased exposure to alternative mobility options. This suggests that companies should prioritize digital engagement strategies, such as in-app incentives, personalized push notifications, and data-driven recommendations to encourage repeat usage. Additionally, customers with high past usage but recent declines should be the focus of targeted reactivation campaigns, leveraging personalized offers based on historical behavior. This study also reveals the impact of seasonal fluctuations, with an increased likelihood of churn during holiday periods. To counteract these effects, companies should implement adaptive retention initiatives, such as time-sensitive promotions, bundled ride discounts, or loyalty rewards that align with seasonal demand shifts.Beyond promotional strategies, improving service quality is critical. Ensuring timely pickups, well-maintained vehicles, courteous drivers, and seamless payment options significantly improves the customer experience and reduces churn. Strengthening customer relationships through effective feedback mechanisms, prompt problem resolution, and high-quality customer support can further increase satisfaction and loyalty. In addition, reactivation campaigns targeting inactive users can be critical. Offering discounts or bonuses to infrequent users can incentivize usage, while tiered loyalty programs encourage consistent patronage through escalating rewards.

By implementing these data-driven strategies, taxi companies can improve customer retention, optimize resource allocation, and gain a competitive advantage in the market. Beyond business-led initiatives, policymakers play a critical role in fostering an environment that encourages customer loyalty and innovation within the taxi industry. Strategic actions such as investing in infrastructure and data access, and prioritizing service quality, safety, and affordability can contribute to long-term industry growth. In addition, to mitigate the impact of seasonal fluctuations in demand, policymakers can work with local tourism boards to promote off-season travel and encourage the use of taxis to explore lesser-known destinations. Increased collaboration between taxi companies, technology providers, and regulators is also essential to developing innovative solutions and addressing emerging industry challenges. By integrating data-driven customer engagement strategies with

supportive policies, the taxi industry can increase customer loyalty, improve service quality, and ensure long-term success.

6. Conclusions and future work

This section outlines the key conclusions drawn from the work conducted (Section 6.1), followed by a discussion of the study's limitations and proposed directions for future research (Section 6.2).

6.1. Main conclusions

This paper contributes to the literature by proposing predictive models to identify customers with a high likelihood of churning in the context of the taxi industry, thereby assisting taxi service providers to improve customer retention. Predictive models were developed using various machine learning techniques, including LightGBM, RF, ANN and a Stacking Ensemble model that leverages the predictions of individual LightGBM, RF, and ANN models as input features. Resorting to a dataset composed by 47 variables that describe the past travel behavior of customers, 2 distinct classification models were built for each technique: one model incorporating all of the variabless and other considering the most impactful variables derived from a greedy feature forward selection process. Furthermore, and driven by a profit indicator, we conducted a customized hyperparameter tuning step to optimize the performance of each model. This way, we determined the predictors and hyperparameter settings that resulted in the most profitable model for each technique. To predict the churning behavior of customers, we investigated variables that quantify the recency and frequency of usage, together with other predictors linked to the customers' satisfaction level towards the taxi service. Among the techniques employed, our results showed a low level of consistency in terms of predictive variables that compose the best performing combinations. Notwithstanding, we concluded that the contrast between what happened in the same month in the previous year and the most recent travel behavior for each customer may suggest a significant behavioral change. Furthermore, traditional holiday months for the general public significantly influence the travel behavior of taxi users. Additionally, the outcomes of our analysis exposed the significant impact of feature selection on model profitability, as the models trained with the optimized set of predictors consistently outperform their full-variable counterparts. On the whole, excepting for the ANN model with best performing variables, all developed models exhibit a superior ability to distinguish churners from non-churners when compared to a baseline model. Accordingly, these can be considered valuable tools in assisting the company's decision-making process. The predictive power of the models was also evaluated when targeting increasing proportions of high-risk churning customers by assessing the respective lift value. All built models have demonstrated the ability to properly classify a higher percentage of churners than a random model, with the greatest difference noted when considering lower proportions of customers. Indeed, when targeting the top 5% of churning customers, RF model, counting on the complete set of variables, correctly identify about 90% more churners than a random model. The primary value of the proposed models lies in their ability to accurately predict churn, enabling companies to target retention efforts effectively and maximize profit by focusing resources on customers at highest churn risk, thus avoiding wasted marketing spend. Moreover, the interpretability analysis carried out revealed key churn indicators, including preference for digital channels (website/app), travel patterns changes due to holidays season and weekends, and high service utilization in the previous year. As an addition to the existing literature in this field, this work demonstrates the pertinence of exploring changes in travel behavior of customer to infer customers' propensity to churn. Furthermore, our findings highlight the large potential of utilizing machine learning techniques, combined with model interpretability analysis, to predict customer churn among taxi users, doing so by leveraging data that are often readily available and accessible to companies.

6.2. Limitations and future work directions

The proposed models have certain limitations as their development relies on historical travel data and certain assumptions, namely those related to the customer selection and churn classification criterion. The model outcomes can be influenced by customers' low service usage, even just a temporary decrease. For future research, we consider that further improvements in the results can be obtained by changing the criteria for selecting the customers included in the analysis and for defining the target variable. Furthermore, we note that the predictive variables explored fall within the bounds of the data provided to us. However, we believe incorporating additional predictive variables can leverage the predictive power of our models. For instance, spatial data such as the pick-up and drop-off locations could be explored. Additionally, including information regarding customer satisfaction with the service collected through a survey could enhance the ability to capture behavioral changes and engagement levels. Furthermore, to have access to the customer identifier, the dataset analysed does not include the services departed from taxi stands or directly hailed by customers in the street, which can misrepresent the travel behavior of taxi users and distort the description made concerning the taxi business in Coimbra. Another limitation of our work is the lack for a more in-depth analysis for investigating the impact of seasonality on churn patterns and explore how model performance may vary over time. Moreover, in this work, fixed parameters values are used for the EMPC profit indicator estimation, which, while based on prior research, must be customized to suit particular real-world contexts. To enhance practicality, the profit indicator should account for individual customer differences. Moreover, we acknowledge the importance of incorporating the potential impact of seasonal effects, particularly during holiday periods, on customer travel patterns. This research could be further extended to account for such effects on the development of models. Finally, we believe that future research should explore the application of uplift modeling in this context, as it has the potential to identify customers most likely to respond positively to retention efforts, leading to more targeted and potentially more profitable retention actions compared to targeting all high-risk customers.

CRediT authorship contribution statement

A.L.D. Loureiro: Writing – original draft, Methodology, Formal analysis, Conceptualization. **V.L. Miguéis:** Writing – review & editing, Validation, Conceptualization. **Álvaro Costa:** Writing – review & editing, Conceptualization. **Michel Ferreira:** Writing – review & editing, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Hyperparameters tuning

Table A.4
Hyperpameters tuned in each data mining technique and correspondent values.

Technique	Parameter	Values
LightGBM	Boosting	gbdt; dart; goss
	Learning rate	0.05; 0.10; 0.15; 0.20; 0.25; 0.30
	Feature fraction	0.4; 0.6; 0.8; 1.0
RF	Mtry, i.e. the number of variables randomly sampled as candidates at each split	1: $\nu^{(*)}$
	Ntree, i.e. number of trees generated	600
ANN	Number of neurons	30; 45; 60; 75
	Learning rate	0.1; 0.3; 0.5
	Decay	0.1; 0.2; 0.3

$^{(*)}$ Total number of predictors within the combination of predictors under evaluation

Appendix B. Predictive variables

Table B.5
Predictive variables explored to predict churners.

Description	notation	Type	Measures
Total number of services	n_services*	numeric	-
Passenger waiting time (in sec)	avg_wait_time*	numeric	average
	max_wait_time*	numeric	maximum
Time elapsed between services (in days)	avg_difference*	numeric	average
	max_difference*	numeric	maximum
Ratio of passenger waiting time and the service duration	avg_wait_duration*	numeric	average
	max_wait_duration*		maximum
Duration of the services	avg_delta_time*	numeric	average
Percentage variation in the number of services recorded in the 2 previous months	delta_n_services	numeric	-
Percentage variation in the passenger waiting time recorded in the 2 previous months	delta_max_wait_time	numeric	-
Percentage variation in the ratio of passenger waiting time and the service duration recorded in the 2 previous months	delta_max_wait_duration	numeric	-
Variation in the number of services recorded in the 2 previous weeks	delta_serv_2_weeks	numeric	-
Time elapsed between the date of the last service of the previous month and the first day of the target month (in days)	serv_recency	numeric	-
Number of services in the homologous month of the previous year	n_serv_prev_homol_month	numeric	-
Trend of the number of services over the 3 previous months	trend_n_services	categorical	-
Trend of the maximum waiting time over the 3 previous months	trend_max_wait_time	categorical	-
Trend of the maximum time elapsed between services over the 3 previous months	trend_max_difference	categorical	-
Trend of the maximum ratio of passenger waiting time and the service duration over the 3 previous months	trend_max_wait_duration	categorical	-
Holiday month 1 for July, August and September; 0 otherwise	holiday_month	categorical	-
Period of the day in which the customer recorded a higher number of services	usual_day_period*	categorical	-
Day of the week in which the customer recorded a higher number of services	usual_day*	categorical	-
Type of day in which the customer recorded a higher number of services 1 for weekend; 0 otherwise	usual_day_type*	categorical	-
Means by which the customer requested a higher number of services	usual_service_subtype*	categorical	-

(*) Note that this variable is measured for each of the 3 months in the characterization period, being further identified by an additional suffix. For example, the use of the _m1 suffix denotes that this variable is related to the first month of the characterization period, while _m2 means that it refers to the second. Accordingly, the use of the _m3 suffix represents a variable measured in the last of these three months.

Appendix C. Best hyperparameters

Table C.6
Best hyperparameters for each predictive model.

Technique	All variables			Best performing variables		
	Boosting	Learning rate	Feature fraction	Boosting	Learning rate	Feature fraction
LightGBM	gbdt	0.05	1	goss	0.2	0.8
	Mtry	Ntree		Mtry	Ntree	
RF	43	600		1	600	
	Number of neurons	Learning rate	Decay	Number of neurons	Learning rate	Decay
ANN	30	0.1	0.3	60	0.1	0.2

Data availability

The data that has been used is confidential.

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